



Impacts of an AI-based *chabot* on college students' after-class review, academic performance, self-efficacy, learning attitude, and motivation

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Accepted: 26 June 2022 / Published online: 1 August 2022
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Abstract

Review strategies after learning new knowledge are essential for students to consolidate the key points, understand the subject content, analyze aspects of the learning topics, and summarize the knowledge content of learning while mastering new knowledge. However, educators have found that students generally have difficulties seeking help when they encounter learning problems. This could significantly affect their after-class review performances. To cope with this problem, an after-class review approach with an AI (Artificial Intelligence)-based chatbot is proposed in this study to provide students with immediate and quality feedback during the learning process. Moreover, a quasi-experiment was conducted to explore students' learning motivation, attitude, and academic performance when using the AI-based chatbot. Participants were two classes of students from a university in Taiwan. One class with 18 students was the experimental group and the other with 20 students was the control group. The experimental group used the AI-based chatbot in the after-class review, while the control group used the conventional after-class review approach. Research results showed that the application of AI-based chatbots in the review process of public health courses could improve students' academic performance, self-efficacy, learning attitude, and motivation. In other words, chatbots could help students become more active in the learning process. It is noted that after students asked questions, providing them with sufficient feedback during the review process could make them feel recognized and help to establish a relaxing and friendly interaction, thereby improving their academic performance.

Keywords Mobile application · Chatbot · Artificial intelligence · After-class review · Public health courses

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Introduction

Review is an important learning stage which aims to help students master what they have learned and to overview the learning content as well as comprehending the key points of the newly learned knowledge to strengthen their learning performances (Shinogaya, 2012). According to constructivism, learning is the result of mental construction, that is, learners assimilate new information into existing knowledge so that they have the opportunity to actively understand it on their own (Bruner, 1979; Naylor & Keogh, 1999; Piaget & Duckworth, 1970; Terwel, 1999). From this perspective, in the after-class review stage, teachers mainly act as learning facilitators and use various strategies to enhance students' learning outcomes.

Scholars have pointed out that without review, students are likely to forget nearly 42% of the learning content within 20 min of the class (Barrouillet et al., 2004). Therefore, timely review after unit teaching is important for knowledge retention. However, educators have found that some students might hesitate to ask questions no matter whether in class or after class owing to various reasons. Asian students are shy about asking questions because they tend to lack confidence, are not good at communicating with others, and are afraid of being ridiculed for asking simple questions (Stowell et al., 2010). This makes it difficult for instructors, during the review process, to provide effective feedback to individual students to resolve their problems. Researchers have pointed out that the quality and quantity of feedback to students in the teaching and learning process is often insufficient (Carless, 2006; Higgins et al., 2002). To enable students to receive quality and instant feedback during the learning process and to enhance their motivation to learn, as well as facilitating their knowledge construction, it is imperative to use modern technology for after-class review.

The popularization of technology has changed teachers' and students' roles, as well as teaching methods. For example, mobile devices are used for collaborative learning and are now common tools in education. They can also be used for online learning and for intense communication (Heflin et al., 2017; Hsieh & Tsai, 2017; Virvou & Alepis, 2005). Regarding learning topics, chatbots stimulate students' learning, communication, problem-solving, creativity, and other kinds of development (Sung et al., 2016). The addition of artificial intelligence also promotes the possibility of individualized learning, allowing teachers to better understand each student's learning process (Hwang et al., 2020), and further design progress courses based on each student's abilities and needs (Kabudi et al., 2021; Timms, 2016).

An artificial intelligence chatbot is a type of software that uses semantic analysis and natural language processing (NLP) to conduct text and voice conversations, understand the prompts given by users, and create an experience of immediate interaction with defined sentences, providing individualized applications (Shah et al., 2016). Chatbots have been used for many different purposes in various fields, including marketing, customer service, tourism, and education (Bhargava et al., 2020; Casillo et al., 2020; Rao et al., 2020; Schmidlen et al., 2019).

The convenience of artificial intelligence (AI) chatbots for timely feedback has aroused the interest of teachers who have also tried to apply them to the field of education, including the earliest language learning. Due to insufficient time in the classroom and limited opportunities for feedback, AI chatbots serve as a great alternative that could assist with anytime, anywhere learning. They provide students with language exercises, and briefly record their feedback. It was found that most students enjoy using chatbots (Fryer &

Carpenter, 2006) as they can easily access chatbots which meet their requirements (Shawar & Atwell, 2007). Chatbots can help provide relaxing and friendly interaction and contribute to a student-centered learning environment (Ochoa & Wise, 2021; Roblyer et al., 2010).

Public health courses mainly teach about health protection, risk control of infectious disease, and environmental hazards. Extended knowledge and practical experience are often required. Research shows that students are often disappointed with the public health courses offered in medical schools and hope to have more public health practitioners with practical experience involved in the curriculum (Blank & McElmurry, 1988; Tyler et al., 2009) to clarify all the possible doubts during the course review process. Many studies have shown that the knowledge in the previous stage is the learning foundation of post-stage knowledge (Lawson, 1983; Peklaj et al., 2015). If the problems that students encounter in class cannot be solved in time, not only could their prior knowledge be poorly mastered, but the new knowledge could not be well learned. Without proper assistance, this problem could become more serious (Binder et al., 2019; Hailikari, Katajavuori, & Lindblom-Ylanne, 2008).

To enable practitioners with hands-on public health experience to be involved in the curriculum, so that students can understand the continuity of theory and practice, an application (app) for professional content was created which provided training through an AI-based chatbot. The main purpose of the present study was to investigate the effects of using this AI-based chatbot for after-class review on college students' learning achievement, motivation, and attitude in the public health curriculum. It was expected that, due to the advantages of an extended dialogue, the interface would offer the ability to deliver immediate responses, give personalized answers, solve problems encountered by students in the review, and provide them with appropriate feedback that would enhance their learning achievement, attitude, as well as self-efficacy, as suggested by several scholars (Beaudry et al., 2019; Bhargava et al., 2020; Fryer et al., 2020; Go & Sundar, 2019).

Literature review

Review and feedback

Review is a learning strategy. Through the process of review, students will repeatedly retrieve the knowledge they have learned, in the hope of retaining it (Butler & Roediger III, 2007; Thompson & Barnett, 1985). The concept is related to psychological short-term and long-term memory, as the operational process of the memory system, from processing stimulus information to encoding, storing, and finally outputting. Short-term memory is also called working memory, and its functions include short-term storage of information. Short-term memory does not retain information for a long period of time and can be easily lost. To form long-term memory, learners need repeated retrieval or need to ask questions about learning multiple times. The process of repeated retrieval of information is key to long-term memory retention (Butler & Roediger III, 2007; Dirks et al., 2015).

After-class review aims to help learners understand what the topic of the lesson content is about and the levels of the topic, have a clear understanding of the important concepts at each level, and be able to make a summary conclusion of the knowledge content (Dunlosky, 2005). Based on new learning, new and old knowledge can be integrated to achieve a better learning outcome. Reviews are conducted not only after learning something, but also before starting to learn new knowledge. Review activities conducted before

new knowledge learning can remind learners of what they have learned in the past and connect it with the learning of new knowledge; review activities after learning new knowledge help learners to master the new knowledge, and effective review strategies will help learners' memory retention (Little & McDaniel, 2015). To allow students to repeat the learning content, examination worksheets can be used to help them understand and master important knowledge. Short answer examinations can be conducted immediately after class to enhance memory retention and promote learning (Butler & Roediger III, 2007). They are also a useful tool for helping students solve problems related to unknown content in class.

Research indicates that prior knowledge is fundamental for learning new knowledge (Lawson, 1983; Peklaj et al., 2015). Clarifying doubts and providing detailed answers during course review tests is also an important part of teaching. Students feel that limited or negative feedback could cause frustration (Ferguson, 2011; Vattøy & Smith, 2019) when problems that are not understood cannot be solved within the allotted time. The consequence is that their existing learning knowledge is not well mastered, which also affects the formation of new knowledge (Binder et al., 2019).

Studies have also shown that insufficient feedback in terms of the quality and quantity provided for students on their learning is common. To allow students to receive appropriate feedback during the learning process, it is important to enhance their learning motivation and knowledge (Carless, 2006; Chen & Hoshower, 2003); therefore, technology-assisted review and appropriate learning feedback are essential tools to enhance efficiency. This research thus used mobile learning artificial intelligence chatbots to assist students' review and learning.

Chatbots in education

A chatbot is a software application that can recognize text or voice and respond (Fryer et al., 2019; Gbenga et al., 2020; Valtolina & Neri, 2021). Chatbots are glorified decision tree hierarchies presented to users, who can interact with them using keyword recognition, input phrases, or just context. Some chatbots further analyze user input using natural language processing technologies to provide quality conversations (Tai & Chen, 2020), or employ machine learning (ML) or other artificial intelligence (AI) technologies to enhance the database and the accuracy of the responses (Nuruzzaman & Hussain, 2020), so as to provide educational support in a friendlier and smarter way.

Chatbots allow students to ask questions and discuss topics so that they have a sense of active participation during the review process, and build relaxing and friendly interactions leading students to deeper discussions and giving personalized answers, thereby achieving a learner-centered environment (Roblyer et al., 2010). Scholars have indicated that the use of chatbots in education can bring benefits and persistent learning, such as enabling learners to go through the content of learning repeatedly, thereby triggering their interest and improving their concentration on learning (Bii et al., 2018; Fu et al., 2020).

Several scholars have indicated that teachers generally do not have sufficient time to address individual students' needs; moreover, students have few opportunities to interact with teachers in class time, not to mention seeking help from teachers when encountering problems after class (Fisher et al., 1981; Shimaya et al., 2021; Slavuj et al., 2017). From the perspective of constructivism, which emphasizes students' responsibility for their own learning, it is important for students to be actively involved in the learning process rather than playing a passive role in learning (Ertmer & Newby, 1993). As a consequence, there

is a need to provide a solution to assist students in completing learning tasks on their own (Lin & Chang, 2020).

Chatbots are such a potential solution for advising people in many fields, such as medicine (Piau et al., 2019; Schmidlen et al., 2019; Srinivas & Sashi Rekha, 2019), product sales (Rajaobelina & Ricard, 2021; Sotolongo & Copulsky, 2018), the service industry (Moriuchi et al., 2020; Sheehan et al., 2020), and education (Fryer et al., 2017). The application of chatbots in medicine is mostly used in clinics, including for chronic disease management. They have great potential for collecting patients' self-care behavior data and can interact with patients personally to educate and improve clinical effects and enhance patients' self-care ability (Abd-Alrazaq et al., 2020; Roque et al., 2021). Chatbots can also assist in mental health management; one study showed that a chatbot offering three chatting styles gained participants' trust in the chatbot, which encouraged their self-disclosure of their personal experiences and feelings. Further, through conversation log analysis, mental health professionals can understand participants' psychological conditions (Vaidyam et al., 2019). A chatbot has also been used in genetic counseling. An AI chatbot advises patients who receive the results of genetic counseling and allows them to send a message to at-risk relatives through the chatbot to inform them of the genetic test results and to encourage them to be tested (Schmidlen et al., 2019).

However, the use of chat robots in medical education is relatively limited at this stage. Also, to enrich knowledge, medical education requires practical experience. It is also hoped that multiple educational methods (Vozenilek et al., 2004) can be used to allow medical students to maintain their interest and enhance their learning through understanding the clinical background and relevance of the knowledge learned in lectures, then to transfer that knowledge and apply it to clinical practice settings, and to make better preparations for optimal patient care. Additionally, few studies have been conducted to analyze the impacts of using chatbots in practical educational settings on students' learning outcomes as well as explaining the factors affecting the effectiveness of using chatbots in school settings or professional training (Zhang et al., 2020).

Therefore, by referring to constructivism and suggestions of several researchers (Smutny & Schreiberova, 2020), in this study, an AI-based chatbot approach was adopted as an intervention for assisting students' after-class review to improve their academic performance, self-efficacy, learning attitude, and motivation. Accordingly, an experiment was implemented in a public health curriculum. The research questions and hypotheses of the study are as follows:

Research Question 1 Does using the AI-based chatbot for public health curriculum review improve students' learning achievement more than conventional after-class review?

Hypothesis 1 When the pretest scores are controlled, the learning achievement of students with AI-based chatbot instruction is improved significantly more than that of students with teacher's instruction.

Research Question 2 Does using the AI-based chatbot for public health curriculum review improve students' self-efficacy more than conventional after-class review?

Hypothesis 2 When the pretest scores are controlled, the self-efficacy of students with AI-based chatbot instruction is improved significantly more than that of students with teacher's instruction.

Research Question 3 Does using the AI-based chatbot for public health curriculum review improve students' learning attitude more than conventional after-class review?

Hypothesis 3 When the pretest scores are controlled, the learning attitude of students with AI-based chatbot instruction is improved significantly more than that of students with teacher's instruction.

Research Question 4 Does using the AI-based chatbot for public health curriculum review improve students' learning motivation more than conventional after-class review?

Hypothesis 4 When the pretest scores are controlled, the learning motivation of students with AI-based chatbot instruction is improved significantly more than that of students with teacher's instruction.

The AI-based chatbot: the Disease Stewardship app

The AI-based chatbot was developed by the Taiwan Center for Disease Control and Prevention and HTC. In this AI-based chatbot, an AI natural language processing platform (i.e., Taiwan Bidirectional Encoder Representations from Transformers, T-BERT), is used for pre-training the chatbot to enable it to read, write, and listen to Mandarin and Taiwanese. T-BERT uses the Transformer network architecture to read passages from left to right and right to left for analyzing text features. The chatbot, which was developed by the HTC (High Tech Computer Corporation) team, uses deep learning models and techniques for processing huge amounts of data. Combined with AI accelerated computing technology, it calculates semantic judgments with an accuracy rate of 93.7%. Such a natural language processing technique has been widely adopted to analyze the semantics of dialogue to enable computer systems to interact with users in free format (Alsentzer et al., 2019; Ouyang & Jiao, 2021).

As shown in Fig. 1, in this chatbot, a dialogue system plays the role of processing human–computer interactive dialogues, where messages are in the form of voice, text, or pictures. Via accurately comprehending users' requests, the chatbot is able to provide quality answers to users by accessing and presenting the database content, which contains data (including detailed information and practical cases) related to 91 infectious diseases in a decision tree structure with more than 3,600 paths from the root to the leaves, as well as periodical statistical data in the “opendata” format.

The chatbot provides two main functions: real-time pushing of important infectious disease messages and automatic responses to learners' questions. Important infectious disease messages are pushed in real-time using a back-end administration website that allows important messages to be pushed to users instantly. Figure 2 shows the interface of the Disease Stewardship app. The chatbot lets learners understand the real-time infectious disease epidemic status and view important information about the status of domestic and international epidemics, where to buy masks, and when to get vaccines. It also automatically provides daily updates on the epidemic status and important infectious disease-related news.

When a learner adds the "Disease Stewardship" app to LINE (an instant communication system developed by Z Holdings Corporation) and begins interacting with it,

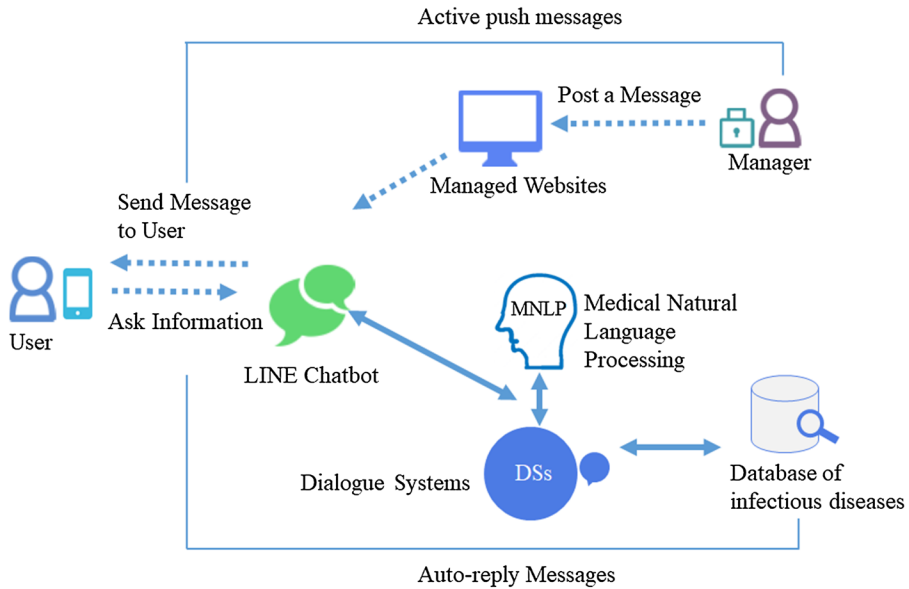


Fig. 1 Structure of the Disease Stewardship App

the Chatbot receives questions from the learner and responds to them. Under the guidance of the review activity worksheet, students can use the app to complete the learning tasks. The following is the learning process:

- Step 1: Case study stage: In this stage, students are asked to analyze practical cases and give answers based on their judgments of the cases, such as “Summer is coming; Xiaohua decided to use consecutive holidays to travel abroad to the islands in Southeast Asia. He had a fever and joint pain when he got back home. He had pain in his posterior eye socket, and after a visit to the doctor he was diagnosed with dengue fever. Xiaohua was very worried and wanted to know more about the symptoms of dengue fever: What should I do if I am infected with dengue fever? How is it treated? What should I pay attention to? Do I need to be isolated?”
- Step 2: System advice stage: In this stage, students are guided to enter the cases into the chatbot to receive advice from it, as shown in Figure 3.
- Step 3: Reflection stage: In this stage, students are guided to compare their answers to the cases with those provided by the chatbot and to identify differences between the answers.
- Step 4: Refinement stage: In this stage, students are asked to learn more about epidemiology based on what they have found in the previous stages. For example, they might need to know more regarding the additional factors causing the diseases and other relevant diseases as well as the distribution of diseases in different areas, as shown in Figure 4.
- Step 5: Conclusion Stage: In this stage, students are asked to summarize what they have learned in the activity.



Fig. 2 Important infectious disease information is pushed in real time



Fig. 3 Illustrative example of the system advice stage



Fig. 4 Illustrative example of the refinement stage

Experimental design

Participants

This research was conducted using a quasi-experimental research method. The participants were two classes of freshmen who took a healthcare course in a medical university located in northern Taiwan. One class consisting of 18 students (2 males and 16 females) was the experimental group, and the other class consisting of 20 students (6 males and 14 females) was the control group. The experimental group used an AI-based chatbot to assist with the course review, while the control group interacted directly with the instructor. All students were taught by the same instructor who had more than 15 years of teaching experience. The instructor was not a member of the research team and was not aware of the purpose of the research.

Instruments

The learning motivation questionnaire from Wang and Chen (2010) was utilized in this study to evaluate students' learning motivation. It consists of six items (e.g., "In a class like this, I prefer course material that really challenges me so I can learn new things") with a 5-point Likert Scale. The Cronbach's alpha value of the questionnaire was .79. Also, a questionnaire proposed by Hwang, Yang, and Wang (2013) was utilized to examine the students' learning attitude, in which seven items (e.g., "I think learning public health courses is interesting and valuable") with a 5-point Likert Scale were included. The Cronbach's alpha value of the questionnaire was .88. In addition, a questionnaire proposed by Pintrich et al., (1991) was utilized to examine the students' self-efficacy, in which eight items (e.g., "I believe I will receive an excellent grade in this class") with a 5-point Likert Scale were included. The Cronbach's alpha value of the scale reached .90.

Administered to obtain students' perceptions of the learning approach they had experienced, a list of interview questions proposed by Hwang et al., (2009) was adopted. These questions are listed as follows:

- Question 1: How is studying and reviewing in this way different from the way you have experienced (or expected) before? For example, how does using chatbots to assist with learning and reviewing compare with your previous learning and reviewing? Does it work? Why or why not?
- Question 2: What advantages do you think this learning method has overall?
- Question 3: For which part do you think you achieved the most in this way? In which part did you learn the most? Please give concrete examples.
- Question 4: What can be improved about this approach (e.g., system features or interface design)? Please give specific examples.
- Question 5: Would you like to have the opportunity to learn in this way again in the future? What are the subjects? Why are these subjects appropriate?
- Question 6: Would you recommend this system or this way of learning to your classmates? Why do you think they need to learn in this way? Or would they prefer to learn in this way?
- Question 7: Would you recommend that teachers use this system or this way of teaching? Why do you think they need to teach in this way? Or would they prefer to teach in this way?

Experimental Procedure

This study was conducted using the quasi-experimental research method. The experiment explored the review activities of students taking a public health infectious disease unit course using a chatbot in the experimental group. The worksheet was reviewed after the class was completed. The experiment was conducted once a week for 2 weeks in a row. There were two sessions each week, each of about 50 min in length. Before the experiment, both groups were tested for their learning achievement, learning motivation, learning attitude, and self-efficacy, and a pre-measurement test was also implemented. The abovementioned experimental procedure is shown in Fig. 5.

The 1st week of the learning course covered infectious disease prevention and treatment, new type A influenza diseases and study review, and the 2nd week covered dengue fever and study review. Both groups received instruction from the same teacher.

Following that, the experimental group used the AI-based chatbot with the worksheet provided by the teacher to review the knowledge of infectious disease prevention, as shown in Fig. 6. If students had questions regarding the AI chatbot or the learning tasks, they could email the teacher with their questions. On the other hand, the control group students adopted a traditional interactive review; that is, they reviewed the learning content and completed the worksheets with the assistance of the teacher by asking the teacher about problems they encountered during the learning process.

After the course was completed, a post-measurement form of learning motivation, learning attitude, and self-efficacy, and a posttest were administered to both groups. Besides, a semi-structured questionnaire was conducted to evaluate experimental group students' views on the learning and review methods of chatbots.

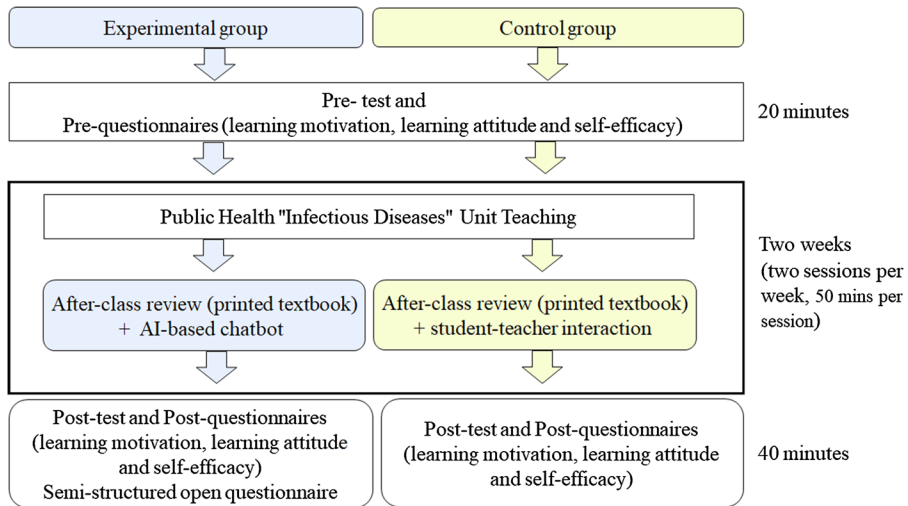


Fig. 5 Experimental procedure

Results

A quasi-experiment was conducted in this research, with the subject being the review process of the infectious diseases unit in a public health course. The experiment lasted for 2 weeks. The experimental group used an AI-based chatbot and printed textbook to review, while the control group used traditional review involving interactions with the teacher and the printed textbook. The experiment conducted a post-test to detect changes in learners' academic performance, learning motivation, learning attitude, self-efficacy, and classroom participation. ANCOVA analysis and qualitative analysis were performed. The results are as follows.



Fig. 6 Students of the experimental group doing after-class review with the chatbot

Analysis of learning achievement

To exclude the influence of students' prior knowledge, the one-way ANCOVA was conducted using the pre-test of learning achievement as a covariate and the post-test of learning achievement as the dependent variable to analyze the learning achievement of the two groups of students. In order to judge whether the use of ANCOVA was proper, the Levene's test of determining homogeneity of variance was not violated ($F=0.09$, $p>0.05$), indicating that the null hypothesis was tenable and the variance was equal across groups. In addition, a test of the homogeneity of regression coefficients within groups was performed. It was found that the assumption of homogeneity was not violated with $F=0.012$ ($p>0.05$), implying that the hypothesis of homogeneity of the regression coefficients was satisfied. Then ANCOVA could be carried out. ANCOVA procedures produce an adjusted mean which represents the average value of the dependent variable considering the covariate and its relationship to the independent variable and the dependent variable (Mishra et al., 2019).

By incorporating an AI-based chatbot in the review process, the experimental group's average academic performance was 66.33, with a standard deviation of 10.39. Comparatively, the control group's average was 57.40, with a standard deviation of 12.33. The adjusted means and standard error of the scores were 65.94 and 2.14 for the experimental group, and 57.75 and 2.03 for the control group. The ANCOVA analysis showed significant differences between the two groups ($F=7.70$, $p<0.01$, $\eta^2=0.18$), with a large effect size ($\eta^2=0.18$) (Cohen, 1988), meaning that the experimental group and control group had different learning results because of the different review methods. This shows that the addition of an AI-based chatbot to the learning review was helpful for students' performance, as shown in Table 1.

Analysis of self-efficacy

To exclude the influence of the pretest scores, one-way ANCOVA was conducted using the pre-test of self-efficacy as a covariate and the post-test of self-efficacy as the dependent variable to analyze the self-efficacy of the two groups of students. In order to judge whether the use of ANCOVA was proper, the Levene's test of determining homogeneity of variance was not violated ($F=2.43$, $p>0.05$), indicating that the null hypothesis was tenable and variance was equal across groups. In order to judge whether the use of ANCOVA was proper, a test of the homogeneity of regression coefficients within groups was performed. It was found that the assumption of homogeneity was not violated with $F=1.871$ ($p>0.05$), implying that the hypothesis of homogeneity of the regression coefficients was satisfied. Then ANCOVA could be directly employed.

Table 1 One-way ANCOVA results of learning achievement

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	Std. error	<i>F</i>	η^2
Experimental	18	66.33	10.39	65.94	2.14	7.70**	0.18
Control	20	57.40	12.33	57.75	2.03		

** $p<0.01$.

Table 2 One-way ANCOVA results of self-efficacy

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	Std. error	<i>F</i>	η^2
Experimental	18	4.15	0.50	4.10	0.11	9.25**	0.21
Control	20	3.61	0.38	3.65	0.10		

** $p < 0.01$.

In terms of learning self-efficacy, the average of the self-efficacy ratings for the experimental group was 4.15, with a standard deviation of 0.50, while the average for the control group was 3.61 with a standard deviation of 0.38. The adjusted means and standard error of the scores were 4.1 and 0.11 for the experimental group, and 3.65 and 0.10 for the control group. The ANCOVA analysis again showed a significant difference ($F=9.60$, $p < 0.01$, $\eta^2=0.21$), with a large effect size ($\eta^2=0.21$) (Cohen, 1988), which means that the experimental group and the control group had different learning self-efficacy due to the different review methods. The experimental group had better self-efficacy due to the additional assistance of the AI-based chatbot in the review process, as shown in Table 2.

Analysis of learning attitude

To exclude the influence of the pretest scores, one-way ANCOVA was conducted using the pre-test of learning attitude as a covariate and the post-test of learning attitude as the dependent variable to analyze the learning attitude of the two groups of students. In order to judge whether the use of ANCOVA was proper, the Levene's test of determining homogeneity of variance was not violated ($F=0.00$, $p > 0.05$), indicating that the null hypothesis was tenable and variance was equal across groups. In order to judge whether the use of ANCOVA was proper, a test of the homogeneity of regression coefficients within groups was performed. It was found that the assumption of homogeneity was not violated with $F=0.130$ ($p > 0.05$), implying that the hypothesis of homogeneity of the regression coefficients was satisfied. Then ANCOVA could be directly employed.

In terms of learning attitude, the average of learning attitude ratings for the experimental group was 4.36, with a standard deviation of 0.59, while the average for the control group was 3.67 with a standard deviation of 0.47. The adjusted means and standard error of the scores were 4.29 and 0.13 for the experimental group, and 3.74 and 0.12 for the control group. The ANCOVA analysis again showed a significant difference ($F=8.60$, $p < 0.01$, $\eta^2=0.197$), with a large effect size ($\eta^2=0.197$) (Cohen, 1988), which means that the experimental group and the control group had different attitudes towards learning due to the different review methods. The experimental group had a better learning attitude due to the addition of an AI-based chatbot in the review process, as shown in Table 3.

Table 3 One-way ANCOVA results of learning attitude

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	Std. error	<i>F</i>	η^2
Experimental	18	4.36	0.59	4.29	0.13	8.60**	0.19
Control	20	3.67	0.47	3.74	0.12		

** $p < 0.01$.

Analysis of learning motivation

To exclude the influence of the pretest scores, one-way ANCOVA was conducted using the pre-test of learning motivation as a covariate and the post-test of learning motivation as the dependent variable to analyze the learning motivation of the two groups of students. In order to judge whether the use of ANCOVA was proper, the Levene's test of determining homogeneity of variance was not violated ($F=2.63$, $p>0.05$), indicating that the null hypothesis was tenable and variance was equal across groups. In order to judge whether the use of ANCOVA was proper, a test of the homogeneity of regression coefficients within groups was performed. It was found that the assumption of homogeneity was not violated with $F=3.084$ ($p>0.05$), implying that the hypothesis of homogeneity of the regression coefficients was satisfied. ANCOVA could be performed.

In terms of learning motivation, the average for the experimental group was 4.22 with the standard deviation being 0.52, while the average for the control group was 3.67 with the standard deviation being 0.36. The adjusted means and standard error of the scores were 4.12 and 0.10 for the experimental group, and 3.77 and 0.09 for the control group. The ANCOVA analysis shows significantly different results for the two groups ($F=5.39$, $p<0.05$, $\eta^2=0.13$), with a medium effect size ($\eta^2=0.133$) (Cohen, 1988), which means that the experimental group and the control group had different learning motivations because of different review methods. For the experimental group, the addition of an AI-based chatbot greatly improved the students' motivation, as shown in Table 4.

Qualitative analysis of students' perceptions

To obtain students' perceptions of the AI-based chatbots in the review process of public health courses, 18 students from the experimental group were interviewed. The interview questions were revised by two experienced experts to confirm their validity, and two independent coders used Cohen's kappa to measure their reliability, with Cohen's kappa 0.9, thus determining inter-rater reliability. As shown in Table 5, students generally perceived the use of AI-based chatbots according to three main dimensions, namely "facilitating learning engagement," "ubiquitous learning," and "personalized learning factors that enhance intrinsic motivation."

In terms of the first positive dimension, "facilitating learning engagement," students generally expressed their interest in using new technology in the classroom to facilitate their learning engagement. For example, SE14 stated that, "There is no time limit on the use of chat robots, so I can easily ask questions at any time," and SE12 indicated that "the chatbot can let us understand the current trend of technology, so that we have more access to learning resources," and SE15 expressed that "the immediate response from the chatbot

Table 4 One-way ANCOVA results of learning motivation

Group	<i>N</i>	Mean	<i>SD</i>	Adjusted mean	Std. error	<i>F</i>	η^2
Experimental	18	4.22	0.52	4.12	0.10	5.39*	0.13
Control	20	3.67	0.36	3.77	0.09		

* $p<0.05$.

Table 5 Summary of the students' opinions

Theme	Code	The number of times mentioned
Facilitating learning engagement	Feeling respected and being more willing to be involved in the conversations	6
	Feeling like interacting with the chatbot to learn more from the rich resources	11
Ubiquitous Learning	Providing accurate answers in a timely manner without being limited by location and time	10
Personalized learning factors that enhance intrinsic motivation	Feeling more willing to learn owing to the provision of immediate answers to individual questions	8
	Feeling eager to learn owing to the helpful information provided to enable knowledge acquisition in an effective way through the simple interface	6

satisfied my need for answers and I felt respected.” In the process of learning, the chatbot can strengthen communication between students and respond to students’ problems promptly.

As for the second positive dimension, “ubiquitous learning,” several students referred to the advantages of using AI chatbots as ubiquitous learning. For example, SE07 stated that “knowledge can be easily obtained through chatbots at any time, hence enhancing my learning review”; SE10 commented that “my questions and needs were fulfilled at any-time, anywhere”; SE13 expressed that “the chatbot can provide answers accurately and in a timely manner. I could inquire at anytime and anywhere.” If students cannot solve their problems immediately, it is impossible to fully understand them, no matter how much new knowledge is given. Therefore, if questions could be answered immediately by chatbots, the learning process would surely be enhanced.

Regarding the perspective of the third positive dimension, “personalized learning factors that enhance intrinsic motivation,” it is the personalized learning factors that enhance intrinsic motivation. For example, SE01 indicated that “using chatbots was different from traditional learning; it helped generate curiosity, was easier to interact with, and I was more willing to ask questions when reviewing.” SE08 commented that “during the review process, I could obtain relevant information and gain new knowledge through simple interface questions and answers, which made me want to learn more.” SE18 expressed that “chatbots could answer individual questions in a timely manner.” It is noted that during the learning process, how to promote students’ engagement and motivation to learn and receive knowledge, continue to strengthen students’ learning motivation, and continuously promote students’ active participation in learning is a vital process.

Students stated that the AI-based chatbot approach was more interactive and enhanced review. It can be deduced that students’ learning outcomes improved because they had the opportunity to learn about the current state of public health knowledge and to think deeply by exploring relevant information. This is one of the goals of the AI-based chatbot approach, that is, to provide individual learners with timely knowledge to solve theoretical and clinical practical problems. Findings are in line with a previous study by Chen and Kuo (2022) in which mobile chatbot-based learning methods were found to stimulate students’ potential and improve their learning outcomes.

On the other hand, students also mentioned the disadvantages of using chatbots. These include, “because a chatbot is not a real person, when no keywords were given, answers like ‘I don’t know’ sometimes pop up.” A chatbot is just a machine and I prefer the traditional way of reviewing homework”; “it’s more innovative and interesting than a traditional paper review, but I’m afraid I didn’t receive the full information that I expected.” Chatbots are still in the developmental stage and sometimes are not able to fully understand the commands given by the user or to deal with more complex issues, so there is a need to create a human experience to grasp the needs of the learner to perform learning review in a more comprehensive and easier manner. To further explore the factors that affect student learning, an AI-based chatbot approach can be used in the future to conduct a more in-depth coding analysis after conversations have been generated.

Discussion and Conclusions

This study applied the AI-based chatbot in the review process of a public health course to understand the effects of chatbots on students' academic performance, self-efficacy, learning attitudes, and motivation during a public health course review.

In terms of the post-class review, students who used the AI-based chatbot learning method significantly outperformed those who used the teacher-student interaction learning method. In terms of learning achievement, the average academic performance of the experimental group was 66.33, which was significantly better than the 57.4 of the control group, with a large effect size. Moreover, in the self-efficacy, learning attitude, and learning motivation aspects, the experimental group performed significantly better than the control group with a medium to large effect size. This could be due to the fact that chatbots are able to provide instant feedback without being limited by location and time; that is, learning problems can be resolved immediately, as indicated by several previous studies (Butler & Roediger III, 2007; Shinogaya, 2012). Researchers have also indicated that students generally prefer to seek help from computer systems rather than from teachers, in particular, Asian students who tend to be shy about asking questions (Stowell et al., 2010). Similar reasons could be applied to the finding of self-efficacy. That is, the chatbot not only helped students understand the continuum of theory and practice, but also provided instant solutions for solving the problems they encountered, which could therefore improve their confidence in learning and practicing. Several researchers have emphasized the importance and effectiveness of providing instant feedback to individual students to increase their self-efficacy as well as their learning performances (Epstein et al., 2002; Ferguson, 2011; Wilson et al., 2021).

With regard to learning attitudes, the chatbot enables students to learn and seek help via an easy-to-use and interactive interface. As suggested by Chen et al. (2020), students are generally more willing to learn in a student-centered learning environment; therefore, chatbot-supported learning has great potential for promoting students' engagement and learning attitudes (Chen et al., 2020). Similarly, provision of personalized learning contexts for after-school review through AI-based chatbots allows students to fully schedule their own learning progress, which enables them to learn in an interactive and enjoyable way (Roblyer et al., 2010), and hence enhances their learning motivation and satisfaction. Additional evidence can be found from the interview results. For example, several students stated that learning with chatbots increased their learning engagement and motivation.

In short, the chatbot helped students become more active in their study and provided them with sufficient feedback during the review process (Belcadhi, 2016), thereby improving their academic performance. Students received feedback during the review process and established relaxing and friendly interactions, triggering deeper discussions, and giving personalized answers. Review activities done before learning new knowledge can help them remember what they have learned in the past, and integrate the new and old knowledge to achieve better learning outcomes. Studies have pointed out that brief immediate feedback is one of the key functions in the learning process (Epstein et al., 2002), and it also promotes active learning, that is, students' active involvement in classroom activities to achieve the learning goals. Moreover, only one student sent e-mails to the teacher to ask about the use of the AI chatbot, implying that students generally had no problem using it.

On the other hand, there are some limitations to the present study that should be noted. Although more students in the experimental group were active in terms of asking the chatbot questions, there were some complex questions that the chatbot could not explain.

Therefore, timely guidance is needed to help students reflect on the content, as well as support from the teacher to stimulate their thinking, such as understanding why the problem was important, and how to apply it to the real world to help learners think about solutions to problems. Moreover, the experimental process was based on a 100 min activity; therefore, generalizations about students' learning attitude would be hard to make based on such a short period. Additionally, it should be noted that the effectiveness of the AI-based chatbot approach could be due to the novelty effect as students had had few such experiences before.

With the continuous evolution of technology, the interactive experience of chatbots in the future is anticipated to be more anthropomorphic, being able to understand emotional expressions and make appropriate responses (Smutny & Schreiberova, 2020), and being more visually personified in the interactive user interface (Go & Sundar, 2019). The humanized conversation and the fluency of communication are the keys that focus on chatbot development in the future. To know more about the effectiveness of AI-based chatbots in educational settings, in addition to after-class review, it is suggested that additional studies can be conducted to examine the impacts of using AI chatbots in the before-class preview stage. It is also suggested that future research can explore the effectiveness of using AI chatbots in educational settings from different perspectives, such as learning engagement, critical thinking, reflective thinking, and problem-solving skills. Combining AI and human on after-class review could also be a possible future research topic. Moreover, understanding the factors that influence learners' performance and perceptions could be crucial for researchers and schoolteachers to reflect on and develop better learning methods. In the near future, we plan to apply this approach to other courses with larger sample sizes and for longer periods of time to further investigate its effectiveness from different angles.

Acknowledgements This study is supported in part by the Ministry of Science and Technology of Taiwan under contract numbers MOST-109-2511-H-011-002-MY3 and MOST-108-2511-H-011-005-MY3.

Declarations

Conflict of interest The authors would like to declare that there is no conflict of interest in this study.

Consent to participant The participants were protected by hiding their personal information during the research process. They knew that the participation was voluntary and they could withdraw from the study at any time.

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